

- 12 The space between two adjacent trusses is called ____
 a. Purlin b. Bay c. knee d. Braces

Q2. Solve the following.

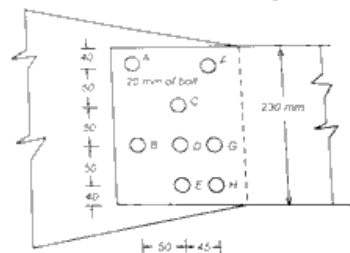
- A) Explain why steel is best suited construction material for the following: 6
 a. Large span industrial truss
 b. Buildings to be shifted
 c. Hilly areas where construction is possible in summer.
 B) Explain the advantages and disadvantages of using steel structures. 6

Q3. Solve the following.

- A) Explain the following terms: 6
 a. Pitch of the bolts
 b. Gauge distance
 c. Edge Distance
 B) Determine the bolt value of 20 mm diameter bolt connecting 10 mm plate in: 6
 a) single shear
 b) double shear.
 Bolt used are 4.6 grade and plate of 410.

Q4. Solve Any Two of the following.

- A) Explain the concept of the shear lag phenomenon with a suitable example 6
 B) Determine the net area of the plate of size 230 mm x 8 mm joined to the gusset plate by 20 mm bolts as shown in fig. 6



- C) Define radius of gyration and slenderness ratio. Also state the maximum values of the slenderness ratio for any two conditions of the compression member 6

Q5. Solve Any Two of the following.

- A) Design a single lacing system for a column composed of 2 ISMC 300 @ 35.8 Kg/m placed back to back at clear spacing of 200mm. Axial factored load on column is 1500KN. Effective length of column is 5m. 6
 B) Explain lacing and battening in detail with neat sketch. 6
 C) Explain with neat sketches 6
 1) Slab base 2) Gusseted Base 3) Grillage Foundation

Q6. Solve Any Two of the following.

- A) A roof truss shed is to be built in Mumbai for an industry. The size of the shed is 6 24m X 6 40m. The height of the building is 12m at the eaves. Determine the basic wind pressure. 6
 B) Describe the process of design of truss with details of 6
 1) Wind load calculations 2) different load combinations
 C) Explain factors affecting plastic moment capacity in detail. 6

*** End ***